Water Automation Master Plan

Pre-plaster conduit schedule + circuit diagrams

WHAT'S INSIDE

HAND THIS DOCUMENT TO

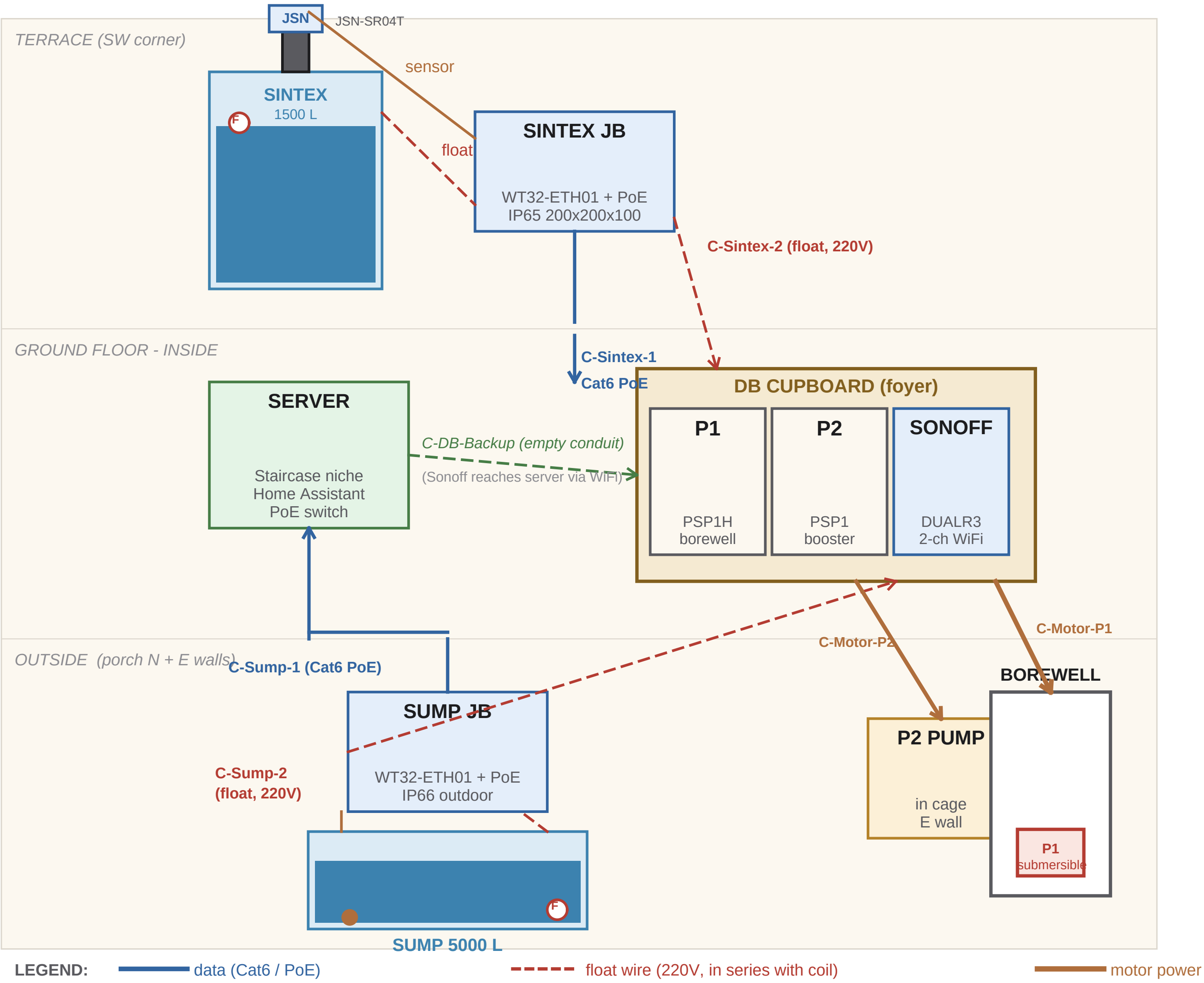
Electrician (pre-plaster) / Plumber (pump piping) / Owner (reference)

Source: electrical/water-automation-conduits.md
Project: Home Interior & Automation - v1.0 - 2026-05-13

System Overview

What's connected to what - pumps, sensors, server, cupboard

Two tanks (Sintex on terrace, sump outside). Two motors (P1 borewell, P2 booster). Both motor starters live inside a lockable cupboard adjacent to the foyer DB.



HOW AUTOMATION FLOWS

Sensors read levels -> JBs send digital values to server via Cat6 PoE -> Server runs Home Assistant logic -> Sonoff in cupboard switches motor contactor coils -> motor runs / stops. Manual override via starter buttons (open cupboard, press green/red). Hardware failsafe via float switches (direct break in coil supply, server-independent).

Master Conduit Schedule

All 7 conduits to lay before plastering

CRITICAL TIMING

Horizontal terrace runs (C-Sintex-1 and C-Sintex-2) MUST be embedded under the terrace screed BEFORE tiling / waterproofing. Cannot retrofit later without breaking tile work.

ID	Size	From -> To	Carries	Length
C-Sintex-1	20 mm	Server niche -> Sintex JB on terrace SW parapet	Cat6 PoE (data + 48V)	~ 42 ft
C-Sintex-2	16 mm	Sintex JB -> DB cupboard	2-core 1.5 mm2 (Sintex float, 220V)	~ 45 ft
C-Sump-1	20 mm	Server niche -> Sump JB on porch W wall (300mm AGL)	Cat6 PoE (data + 48V)	~ 25 ft
C-Sump-2	16 mm	Sump JB -> DB cupboard	2-core 1.5 mm2 (Sump float, 220V)	~ 25 ft
C-DB-Backup	20 mm	Server niche -> DB cupboard	EMPTY (pull string - future Cat6)	~ 12 ft
C-Motor-P1	25 mm	DB cupboard -> Borewell head (outside)	3-core 4 mm2 armoured	depends
C-Motor-P2	25 mm	DB cupboard -> P2 cage on east outside wall	3-core 2.5 mm2 PVC	~ 5 ft

FIVE RULES

(1) Data and mains conduits NEVER share. (2) Parallel data + mains separated by >= 50 mm. (3) Every conduit gets a pull-string left inside. (4) Horizontal terrace runs done before tiling. (5) All JBs IP65 minimum; outdoor JBs IP66.

Per-Conduit Routing

Detail for each of the 7 runs

C-Sintex-1 - Server -> Sintex JB

Route	Vertical through staircase shaft to terrace level; horizontal embedded in terrace screed (under tile bed) to SW corner; rise up parapet to JB at ~ 1.2 m AGL.
Cable / Notes	Outdoor-rated LSZH Cat6. Slab penetration sealed with fire mastic before waterproofing layer.

C-Sintex-2 - Sintex JB -> DB cupboard

Route	Parallel to C-Sintex-1, 50 mm separated. Through screed -> staircase shaft -> horizontal in chase to DB cupboard.
Cable / Notes	2-core 1.5 mm2 double-insulated 220V cable. Mains-class. Separate from Cat6.

C-Sump-1 - Server -> Sump JB

Route	Horizontal in ceiling chase to foyer NW corner; sleeved penetration through porch W wall; vertical drop on outside face to JB at 300 mm AGL, offset ~100 mm NORTH of manhole edge (so cover swing doesn't strike the JB).
Cable / Notes	Outdoor-rated LSZH Cat6. Wall penetration sealed and sloped outward.

C-Sump-2 - Sump JB -> DB cupboard

Route	Parallel to C-Sump-1 outdoors, separate sleeved penetration through porch W wall, indoor chase from foyer NW corner to DB cupboard.
Cable / Notes	2-core 1.5 mm2 double-insulated 220V cable.

C-DB-Backup - Server -> DB cupboard

Route	Through ceiling chase to DB cupboard at standard switch height. Pull string ONLY, no cable.
Cable / Notes	Future Cat6 if ESP32 motor control is wanted later. Cap both ends, label clearly.

C-Motor-P1 - DB cupboard -> Borewell

Route	Exit DB cupboard at floor -> through east wall -> outdoor buried conduit to borewell head -> submersible cable continues in casing.
Cable / Notes	3-core 4 mm2 XLPE armoured. Verify existing run if any. Motor earth pit dedicated.

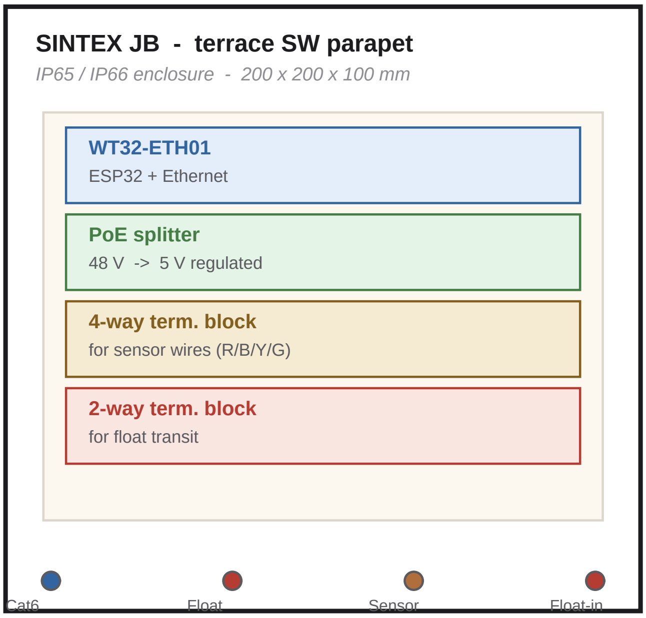
C-Motor-P2 - DB cupboard -> P2 cage

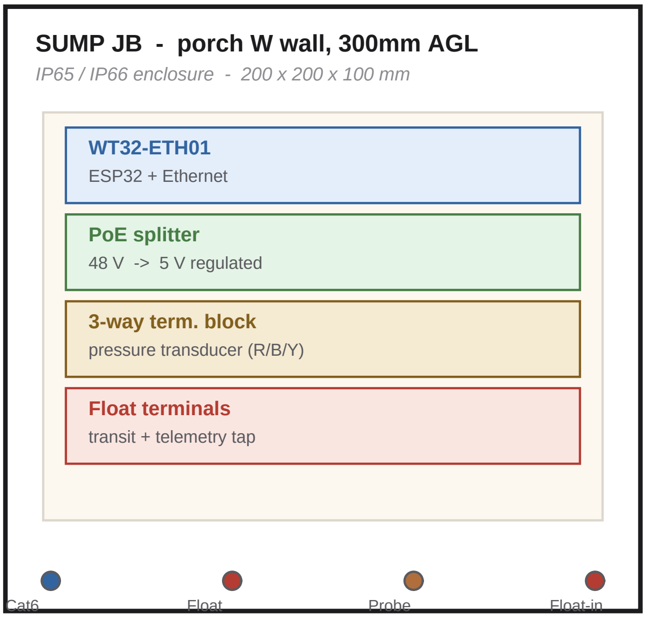
Route	Exit DB cupboard at floor -> through east wall -> short run to steel cage on east wall.
Cable / Notes	3-core 2.5 mm2 PVC double-insulated. Short distance, no armouring needed above grade.

Junction Boxes

Two identical JBs, one per tank

Each JB houses an ESP32 (WT32-ETH01), a PoE splitter, terminal blocks, and acts as the cable transition point between indoor conduit and tank-side cabling.

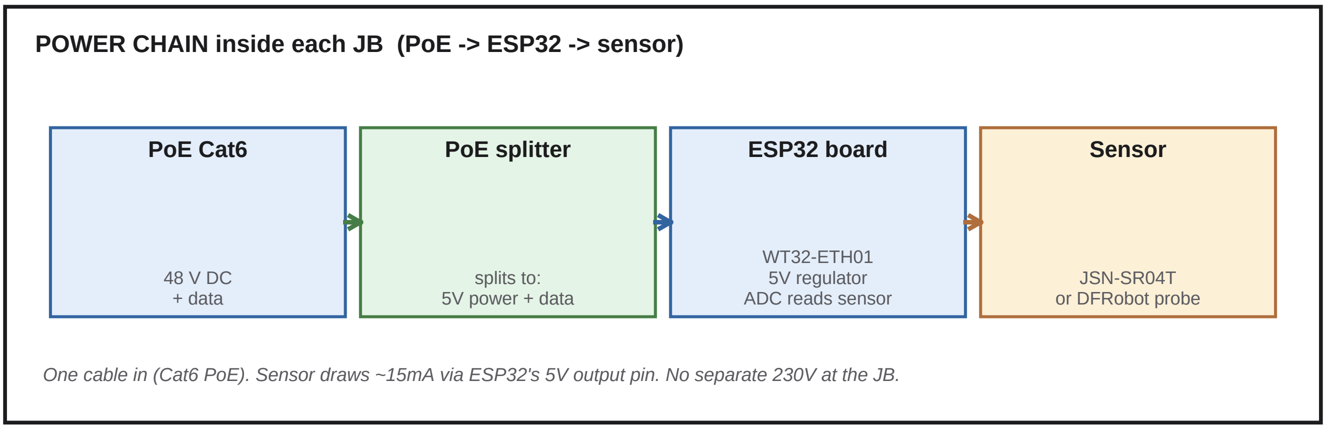




MOUNTING NOTES

Sintex JB: mount on parapet, shaded side. Add fibre-cement sun shield if direct afternoon sun. Sump JB: mount on porch W wall at 300 mm AGL, offset ~100 mm NORTH of manhole edge (cover hinges on W and lies flat against this wall when open). Below corner-window sill. Porch overhang provides shade. Both: gland the bottom face only (top stays clean).

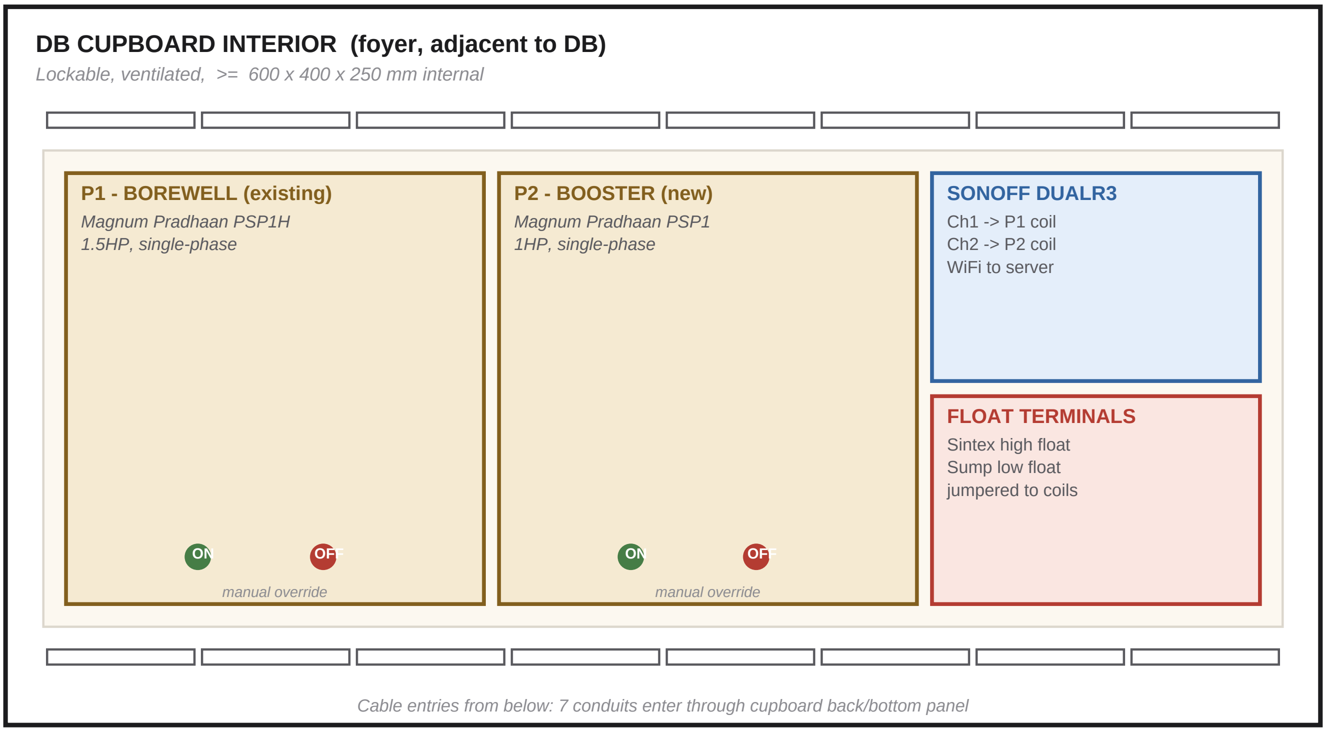
Power chain inside each JB



DB Cupboard

Where both motor starters live, plus Sonoff and float terminals

Lockable wooden cupboard adjacent to the existing DB. Ventilated top + bottom (starters dissipate heat). All 7 conduits enter from below.



Why starters inside the home

Moving both starters from outside (original plan) to inside (final plan) means: (a) manual operation does not need going outside; (b) HOA switches no longer needed (starter green/red buttons serve as manual override); (c) float wiring terminates in one place where both starters can be jumpered in series.

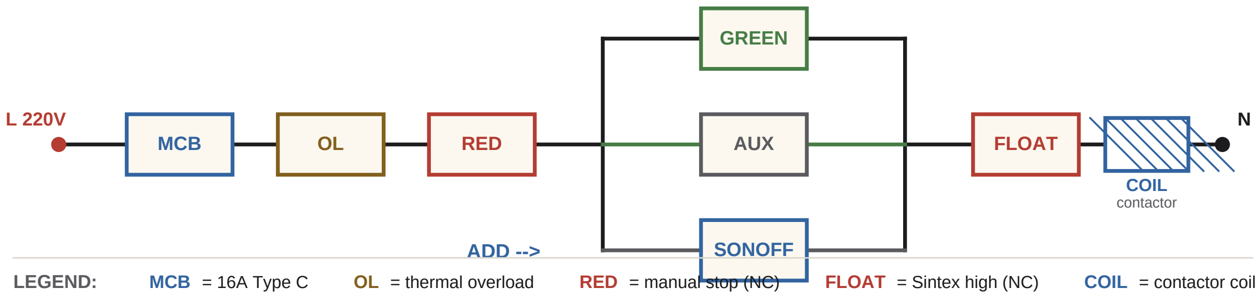
Starter Control Circuit

How the Sonoff taps in without breaking manual operation

Identical pattern for both P1 and P2. The Sonoff channel is paralleled across the Green button + Aux latching contact. Manual buttons keep working; Sonoff can start/stop independently.

STARTER CONTROL CIRCUIT (Magnum Pradhaan + Sonoff tap)

Identical pattern for P1 and P2. Manual + automated paths coexist.



LEGEND: **MCB** = 16A Type C **OL** = thermal overload **RED** = manual stop (NC) **FLOAT** = Sintex high (NC) **COIL** = contactor coil **GREEN** = manual start button (NO) **AUX** = latching contact (NO) **SONOFF** = server start/stop (NO)

Existing path: L -> MCB -> OL -> RED(NC) -> [GREEN(NO) latched by AUX] -> FLOAT(NC) -> COIL -> N
Float failsafe: Sintex high-level float opens (when full) -> coil drops -> motor stops, mechanically.

WHY THIS DESIGN

(1) Manual override preserved (green/red buttons work normally). (2) Thermal overload preserved (we don't bypass it). (3) Hardware failsafe preserved (Sintex high float in series with coil). (4) Server can start/stop via Sonoff channel. (5) Sonoff failure: motor stops, manual still works. Failsafe.

WIRING RULE

The Sonoff's switched output goes in PARALLEL with the green button. Never break the existing manual circuit; only add a parallel path. Test manual operation BEFORE energising the Sonoff for the first time.

Materials Checklist

Buy in advance so plaster doesn't wait on parts

Conduits + accessories

Item	Qty	Notes
20 mm grey PVC conduit	~ 80 m	C-Sintex-1 (13 m) + C-Sump-1 (8 m) + C-DB-Backup (4 m) +
16 mm PVC conduit	~ 50 m	C-Sintex-2 (14 m) + C-Sump-2 (8 m) + spare
25 mm blue PVC conduit	~ 30 m	C-Motor-P1 + C-Motor-P2 + spare
Conduit bends, couplings, junctions	as needed	Same brand as conduit
Pull string (nylon twine)	200 m	Leave in every conduit
Slab + wall penetration sleeves	4	+ fire-mastic for sealing

Cables

Cable	Qty	Notes
Outdoor LSZH Cat6 UTP	~ 50 m	UV-resistant outer jacket
2-core 1.5 mm2 220V flexible	~ 45 m	Double-insulated
3-core 4 mm2 XLPE armoured	?	C-Motor-P1; verify existing first
3-core 2.5 mm2 PVC	~ 5 m	C-Motor-P2 + spare

Junction boxes + glands

Item	Qty
200 x 200 x 100 IP65 / IP66 enclosure	2
20 mm cable gland (for Cat6 entry)	2
16 mm cable gland (for float transit)	2
12 mm cable gland (sensor + float tank exits)	4
Fibre cement sun shield (if Sintex JB direct sun)	1

DB cupboard contents

Item	Qty	Notes
Magnum Pradhaan PSP1 starter (1HP single-phase)	1	New for P2; ~ Rs 3000
Sonoff DUALR3 (Wi-Fi smart switch, 2 ch)	1	~ Rs 1500; flash with Tasmota
4-way 220V terminal block	1	For float wire junction
DIN rail (if cupboard is large enough)	1	Optional, for cleaner mounting
Earthing busbar	1	All motor earths terminate here

Sequencing

Order of operations - who does what, when

Step	Task	Depends on
1	Mark conduit routes on walls (chalk / paint)	-
2	Chase wall channels for conduit	Step 1
3	Lay all 7 conduits + pull strings	Step 2
4	Drill terrace slab + east wall penetrations; install sleeves	Step 3
5	Mount JB backplates (Sintex, Sump)	Step 4
6	Plastering team takes over	1-5 complete
7	Terrace tiling / waterproofing (with C-Sintex-1/2 already embedded)	4 + 6
8	Paint completion	Step 7
9	Install JBs on backplates, connect glands	Step 8
10	Pull Cat6 + float cables through conduits	Step 9
11	Mount DB cupboard + install starters + Sonoff	Step 9
12	Install sensors (ultrasonic in Sintex riser, probe in sump) + floats	Step 10
13	Wire and test	Step 12

Electrician Acceptance Checklist

Sign-off BEFORE plastering proceeds

Walk the site with the electrician. Tick each box only after physical verification. Sign at the bottom.

- [ ] All 7 conduits routed per the master table
- [ ] All conduits have pull strings left inside
- [ ] Data conduits (20 mm) and mains conduits (16/25 mm) separated by >= 50 mm in parallel runs
- [ ] No data and mains cables sharing the same conduit anywhere
- [ ] Slab + external wall penetrations sleeved and marked
- [ ] JB mounting points identified and backplates fixed (Sintex parapet + porch W wall for sump)
- [ ] DB cupboard cable entry positions match the 7 conduit endpoints
- [ ] Conduit ends sealed with caps to prevent debris during plastering
- [ ] Both ends of each conduit labelled with conduit ID (C-Sintex-1, C-Sump-2, etc.)
- [ ] Conduit colour code matches scheme: grey = data, blue = motor power, white = float
- [ ] Photos taken of each conduit run before plastering (for future reference)
- [ ] Owner has reviewed the routes on-site and approved

SIGN-OFF

Electrician

Name + signature + date

Owner

Name + signature + date

Mason (plastering)

Acknowledges conduits in place